

Yaohui Shu

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SUMMARY

Ph.D. candidate in Applied Mathematics and Statistics specializing in stochastic modeling, large-scale simulation, and causal inference for complex systems. Formulates mechanistic models of systems whose governing behavior is uncertain, derives analytical results where they exist, validates them against Monte Carlo evidence, and delivers the analysis as tested, documented software. Same method on turbulent cloud physics (first-author paper in *Geophysical Research Letters*), causal inference in high-dimensional nonlinear systems, and market data.

SKILLS

Programming	Python (NumPy, SciPy, pandas, statsmodels, Numba, PyArrow, TensorFlow), C++, MATLAB, R; Git, pytest, Docker, Linux, SLURM/HPC
Modeling & Simulation	Stochastic differential equations, Langevin and Fokker-Planck models, Ornstein-Uhlenbeck processes, numerical integrators, Monte Carlo simulation at scale, maximum entropy methods
Causal Inference	Causal discovery and graphical models (FCI, LPCMCI), conditional independence testing (kNN, Gaussian copula), transfer entropy, information flow, doubly robust estimation (DML, AIPW, TMLE)
Statistics & ML	Time series analysis, change-point detection, bootstrap and surrogate methods, multiple-testing control, LSTM, XGBoost, PCA, clustering, out-of-sample validation without leakage
Quantitative Finance	Derivatives pricing and Greeks, options positioning analytics, strategy backtesting with transaction costs, financial econometrics, risk management; market-data APIs and columnar (parquet) data stores

EXPERIENCE

Stony Brook University

Teaching Assistant, AMS 595 / DCS 525 Fundamentals of Computing

Stony Brook, NY

Aug 2026 – Present

- Write and deliver the live coding demonstrations for a project-based graduate course in MATLAB, Python and C/C++: scripting, data structures, object-oriented programming, scientific computing, debugging, profiling, version control and build tools.
- Run weekly office hours and support students through the projects that carry the entire course grade.

Brookhaven National Laboratory

Research Assistant: Stochastic Modeling, Simulation & Causal Inference

Upton, NY

Jan 2024 – Jan 2026

Doctoral research, Stony Brook University (ongoing)

Aug 2023 – Present

- Sole doctoral researcher on a DOE-funded program with Stony Brook University: set the research direction, built the models and the code base, and authored the resulting publications.
- Dissertation program: a general method for determining which process drives which in systems that are high-dimensional and nonlinear, contain feedback, and leave the main drivers unmeasured, using analytically solvable systems as ground truth.
- Identified the regime where the standard white-noise approximation fails and replaced it with a finite-memory (Ornstein-Uhlenbeck) formulation that recovers the classical limit and quantifies how memory and noise intensity drive dispersion.
- Converted a solvable stochastic model into an exact benchmark for causal inference: derived time-resolved information flow in closed form and built an ensemble estimator that recovers causal direction where correlation is uninformative; published in *Geophysical Research Letters*.
- Reconciled two competing descriptions of the same system, maximum-entropy statistics and Langevin dynamics, by tracing the discrepancy to the choice of state variable, then generalized the model to any conserved moment order.
- Built `ecausal`, an installable Python package and command-line tool that returns a causal graph with orientation evidence, a linearity verdict and a strength estimate per edge: nonparametric and copula independence tests, bootstrap intervals, multiple-testing control, distributed execution on SLURM clusters, and a test suite pinned to closed-form results.

EDUCATION

Stony Brook University, Ph.D., Applied Mathematics & Statistics (GPA 3.78 / 4.0)

Expected May 2027

Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*

University of Rochester, M.S., Finance; CQF Institute, Certificate in Quantitative Finance

Aug 2021 – Aug 2022

PUBLICATIONS & MANUSCRIPTS

- **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., Li, X. (2026). Stochastic Condensation with Information Flow for Causality Analysis. *Geophysical Research Letters*, 53(17), e2026GL123056. doi:10.1029/2026GL123056
- **Shu, Y.**, Liu, Y., Li, X. A Langevin-Type Stochastic Condensation Model to Explain the Size Distribution based on Maximum Entropy Principle. In preparation for *Physical Review E*.
- **Shu, Y.** Entropy-Based Causal Discovery for Spatiotemporal Dynamical Systems with Feedback and Unmeasured Drivers. Methodology working paper; one further manuscript in preparation.